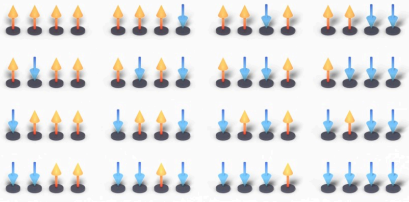
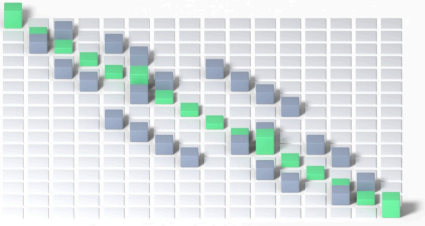
(a)



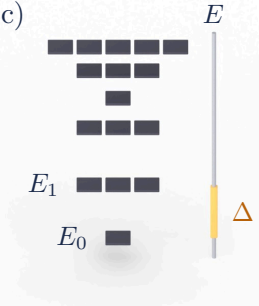
the $2^4 = 16$ product states
$|\sigma_1\sigma_2\sigma_3\sigma_4\rangle$

(b)



$S_i^z S_{i+1}^z$ on the diagonal
$\frac{1}{2}(S_i^+ S_{i+1}^- + \text{h.c.})$ off it

(c)



the 16 eigenvalues